

Portfolio Appendix -examples

Francis Tai

Manufacturing & Operations • Technology Strategy Cross-Functional Leadership • Stakeholder Engagement

Note: This public version presents selected examples of my manufacturing, operations, technology, OEM/ODM, supplier, and cross-functional industry engagements. The full Portfolio Appendix contains additional supporting records and is available upon request.

Delivered business, manufacturing, and OEM/ODM programs across Hi-Tech, Wai Kai, Ngai Lik, Pine/Moosik, and Dale, working with teams across 5+ countries and multiple BUs. These experiences demonstrate negotiation, problem-solving, cross-functional leadership, stakeholder engagement, production ramp-up, and commercial execution.

My professional contacts and friendships over 20 years include Nicholas von Fuerstenberg (CEO, SM/Skymaster, a Thomson Multimedia subsidiary), Steve Cavayero (Founder & CEO, Lafayette Electronics Inc.), and German entrepreneur Lars Windhorst, whom I have known since 1994, when he was 18 and establishing his Asia operations. These contacts are also in my LinkedIn network.

SONY – CD / CD Changer / CDG Player Cost-Down & Technology Transfer

Served as designer and project coordinator, working with multiple Sony engineering teams and business units on CD Player / CD Changer and CDG Player development, cost-down, hardware and firmware integration, and technology transfer at Hi-Tech Precision Products Ltd.



Note: When firmware support was delayed, I contacted Sony's Hong Kong R&D head to identify the responsible engineer and department, then called the relevant contacts in Hong Kong and Japan.

Semiconductor Ecosystem • Global Technology Partnerships • Supplier Development • Product Commercialization

Note: During my first engineering role, I worked directly with Sony engineering on firmware testing and production release, and with its Japan engineering teams on the reference mechanism and new driver-IC demo board. Project progress was reported to company and Group management for this first OEM transfer program, planned for 300,000+ units and multi-year production.

SONY – DVD MPEG + Mechanism Total Solution

Coordinated with Sony's engineering and semiconductor teams on total DVD player solutions, covering MPEG chipset, DVD mechanism, hardware and firmware, forecasting, pricing, RMA, and technical support for Ngai Lik / Denca's OEM/ODM programs.



Semiconductor Ecosystem • Global Technology Partnerships • Supplier Development • Product Commercialization

Note: Across Sony CD/DVD and SANYO programs, I directly engaged department heads when critical issues or delays threatened program execution, reflecting a consistent hands-on escalation approach to maintain technical progress, schedules, and production readiness.

SANYO – Portable VCD Player Mechanism

Led portable VCD player development and commercialization, integrating a SANYO mechanism with a Panasonic MPEG module and coordinating engineering, supply chain, manufacturing readiness, and OEM execution at Wai Kai.



Note: I personally traveled to Shanghai to engage Panasonic's Japan BU Team Lead and its distributor on custom firmware support and module supply.

For urgent matters, I also communicated directly with SANYO Business Development Director Yasuhito Sakurai via his personal email.

Cross-Functional Manufacturing & Quality Network – Tiger Electronics / Hasbro

Representative professional contacts across Accounting, Production Control, Project Management, Shipping, and Quality functions, demonstrating cross-functional engagement within the electronics manufacturing industry.

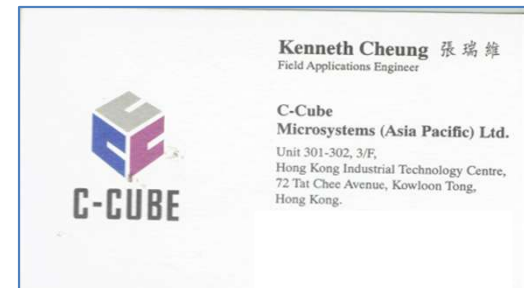


Tiger Electronics, a leading U.S. electronic toy and game company known for Furby and Giga Pets, was acquired by Hasbro in 1998 and operated within the Hasbro group during this engagement."

Note: Worked with client engineering and cross-functional teams to support production ramp-up, tooling, EN71, ISO 9001 compliance, quality, and cost control. Took an LLC Accounting course and Accounting for Engineers in my MSc to support engagement with the client's Accounting team.

C-Cube – DVD Player MPEG

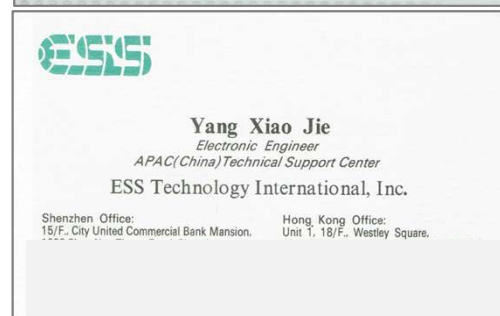
Led development of Ngai Lik / Denca's first DVD player platform, integrating a C-Cube MPEG chipset with a DVS DVD mechanism to support the company's transition from VCD to DVD products, margin improvement, and business growth.



Note: I personally visited C-Cube to work with their engineering team on DVD/DVS firmware improvements, and with sales and management on chipset allocation, pricing, and alignment with DVS production readiness.

ESS Technology – DVD Player MPEG

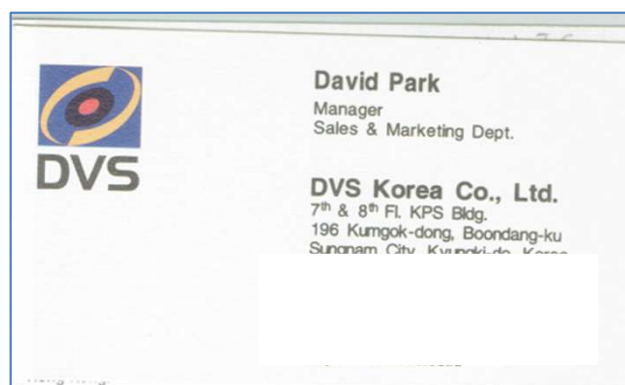
Collaborated with ESS Technology on subsequent DVD player platform development at Ngai Lik / Denca, integrating ESS MPEG chipsets with a DVS DVD mechanism for OEM/ODM product development and commercialization.



Note: My direct technical visits and R&D background enabled faster execution as Sales & Program Lead, allowing me to address technical issues directly and drive program deadlines and production planning without relying solely on engineering reports.

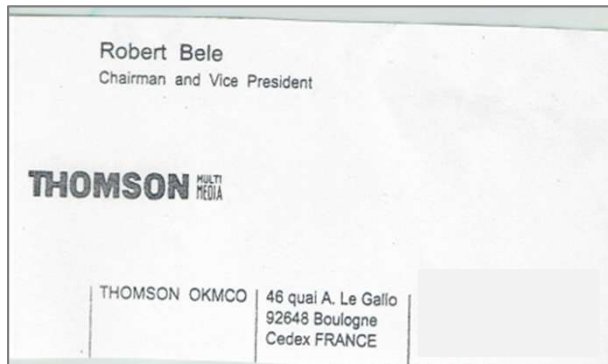
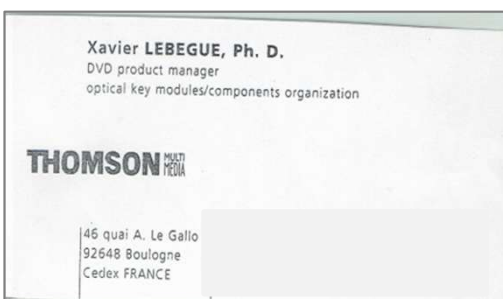
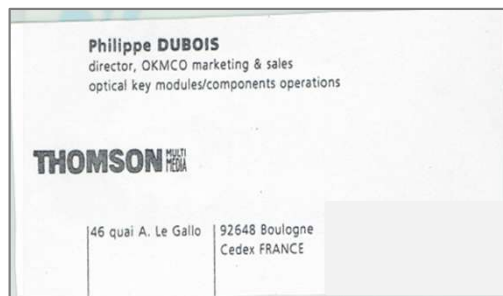
DVS Korea – DVD Player Mechanism

Liaised with DVS Korea, a strategic DVD mechanism supplier, coordinating mechanism integration, hardware and firmware, technical support, material planning, supply readiness, and cross-functional coordination for Ngai Lik / Denca's OEM/ODM DVD player programs.



Note: I requested direct telephone numbers for urgent contact and followed up by phone or in person whenever delays arose.

Executive Meetings for DVD Mechanism Second-Source Evaluation During DVS Shortage



Engaged with Thomson Multimedia and Sharp to evaluate alternative DVD chipset and optical mechanism solutions during the DVS mechanism shortage, covering technical capability, firmware compatibility, cost, and supply readiness. These alternatives were evaluated but not selected for production.

Note: Thomson Multimedia had approximately 73,000 employees worldwide during 2000–2001. I initiated and personally conducted the visit for direct engagement and solution evaluation.

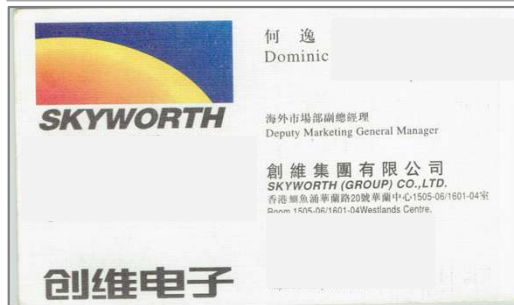
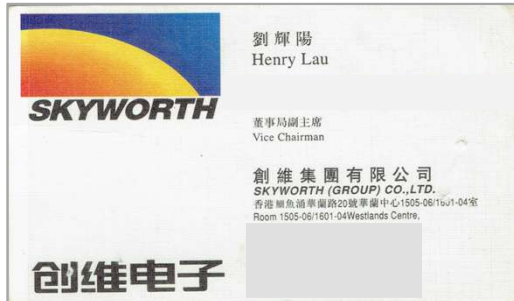


Executive Engagement with Public-Listed Competitors During Critical Supply Shortage

At Ngai Lik /Denca Limited

(HK Stock Code: 00751)

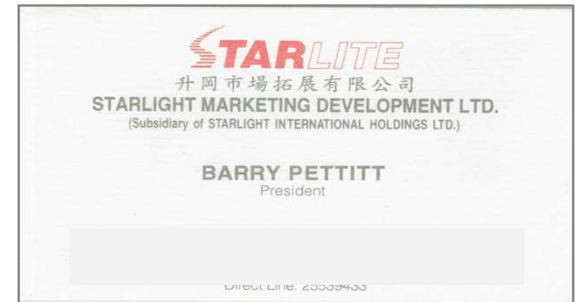
Note: I recorded factory contact details on business cards; these handwritten notes are covered here.



Rapid DVD commercialization, competitive pricing across multiple platform options, and unexpectedly high peak-season demand created a critical DVD mechanism shortage during a period of rapid market price changes. Through executive engagement with Skyworth and other industry competitors, explored potential excess inventory and identified available DVD mechanism stock at Skyworth. Led commercial negotiations to secure approximately 50,000 mechanisms, protecting recurring 100,000-unit L/C shipments for Philco, Lafayette, and SM/Skymaster, maintaining production continuity, and avoiding potential penalties and emergency airfreight costs.

Note: Learned through early engagement with Sony/SANYO global teams, key BU leaders, Hi-Tech management, and the Group Chairman (one of Hong Kong's largest manufacturers), developing executive engagement, cross-functional collaboration, and problem-solving skills.

(HK Stock Code: 00403)



Executive Supplier Engagement • Strategic Sourcing • Supply Chain Risk Mitigation • Alternative Source Development • Production Continuity • OEM Manufacturing

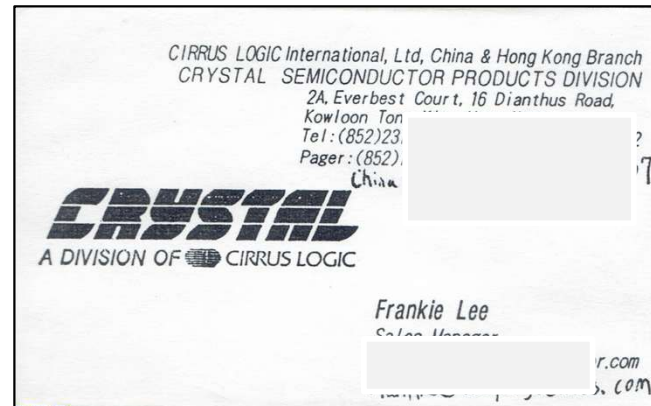
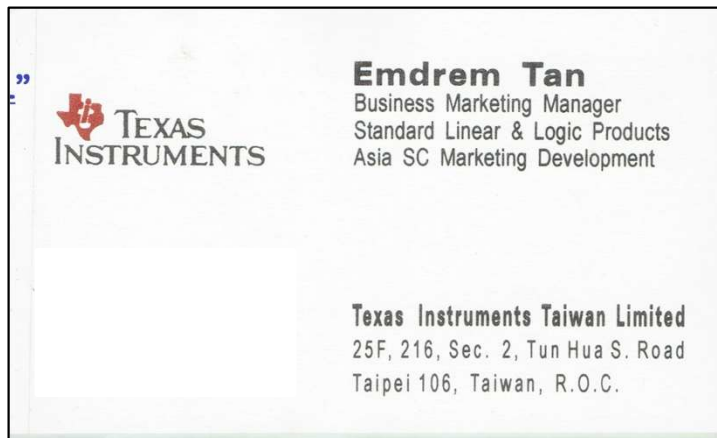
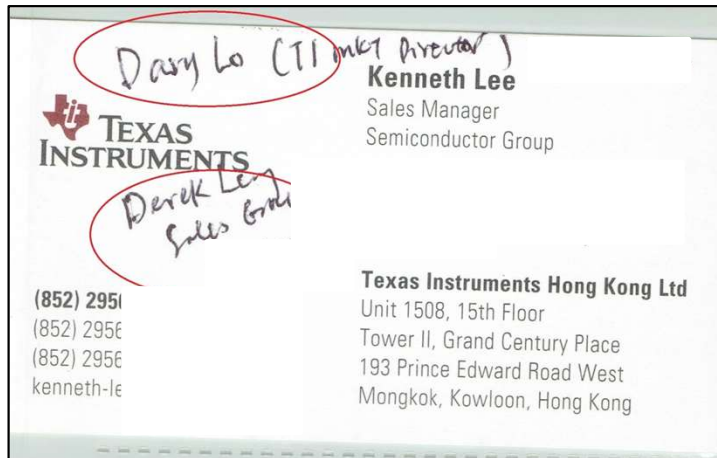


At Pine Technology, evaluated MP3 decoder and DSP platforms from Texas Instruments, Cirrus Logic, and SigmaTel, assessing cost, technical capability, firmware, supplier support, and time-to-market. Selected SigmaTel as the strategic chipset platform to support BOM optimization and OEM product commercialization.

Texas Instruments vs Cirrus Logic vs SigmaTel – MP3 Platform Evaluation | Part 2

At Pine/Moosik Ltd.

Note: I asked TI's sales team to identify and directly contact the Marketing Director and Sales GM for approval of lower pricing on larger orders to support the recovery program.



- Semiconductor Strategy
- Chipset Evaluation
- Platform Selection
- Cost Optimization
- Supplier Collaboration

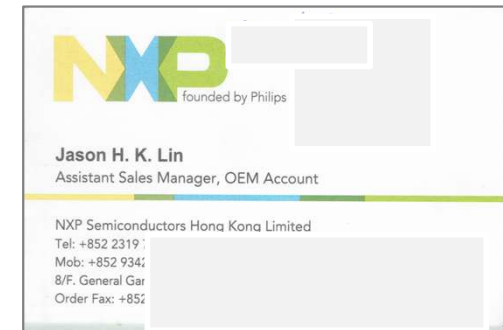
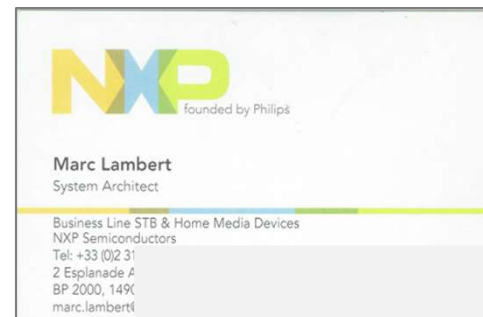
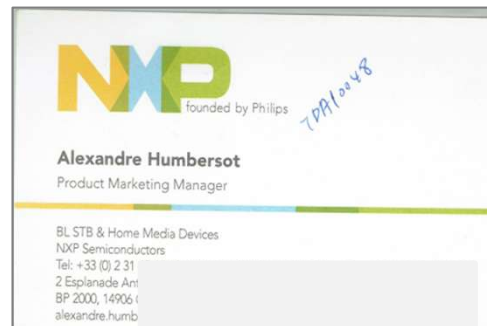
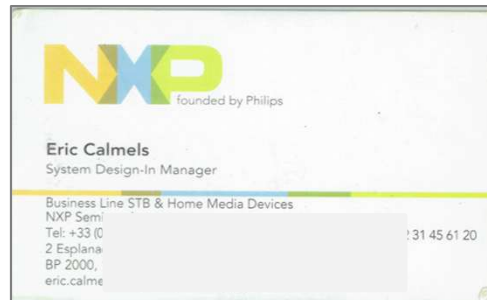
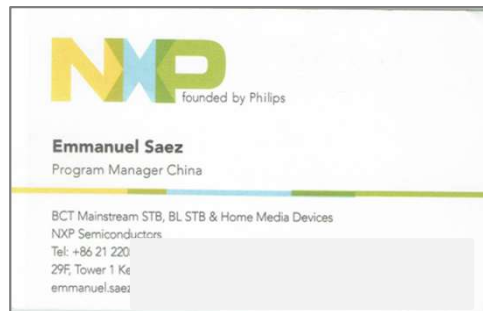
iWedia – DVB-T Receiver Firmware Provider

Collaborated with iWedia on custom DVB-T receiver firmware and software integration for Dale's OEM/ODM programs, supporting Philips-based chipset platform development and commercialization.



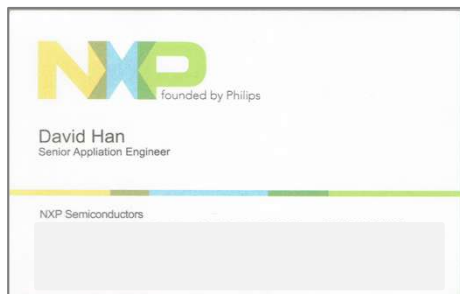
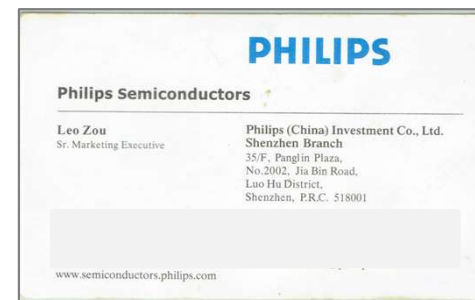
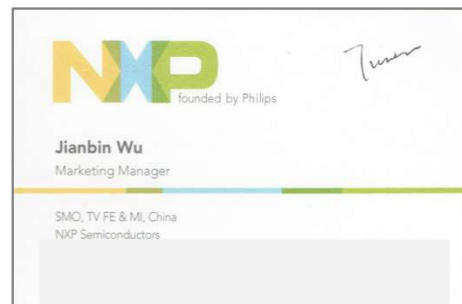
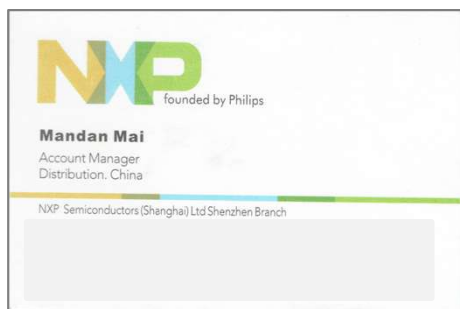
Philips Semiconductors / NXP – DVB-T Receiver | Part -1

Collaborated with Philips Semiconductors / NXP on chipset selection, RF tuners, technical support, application engineering, product roadmaps, custom firmware development, and commercialization for Dale's OEM/ODM DVB-T receiver programs.



Philips Semiconductors / NXP – DVB-T Receiver | Part -2

Note: Repeatedly convinced semiconductor partners to provide custom firmware support despite limited OEM/ODM reference projects, first at Wai Kai and later at Dale for SoC-based DVB-T OEM/ODM programs, differentiating our approach from listed manufacturers using RF tuner modules.



Edge AI - Experimental Validation

Acoustic Feasibility

- Filtering (bandpass + high-pass) improves signal clarity
- Detection achievable under controlled conditions

Noise Robustness

- Traffic noise: stable detection ($\Delta \approx 10\text{--}11.5\text{ dB}$)
- Rain / broadband noise: severe masking ($\Delta < 6\text{ dB}$)

Adaptive Filtering

- Narrow band improves separation but weakens signal
- No single filter works across environments

Technical Uncertainty

- Can acoustic detection remain reliable under real-world noise ?
- Can filtering alone achieve consistent performance ?
- What Δ threshold is required for edge deployment ?

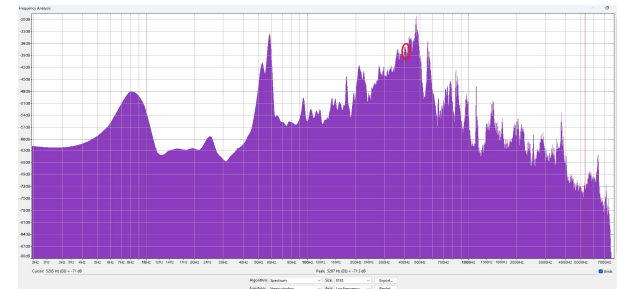
Key Learning

Acoustic-only detection is unreliable in real-world noise

→ Multi-sensor fusion (audio + vision) is required

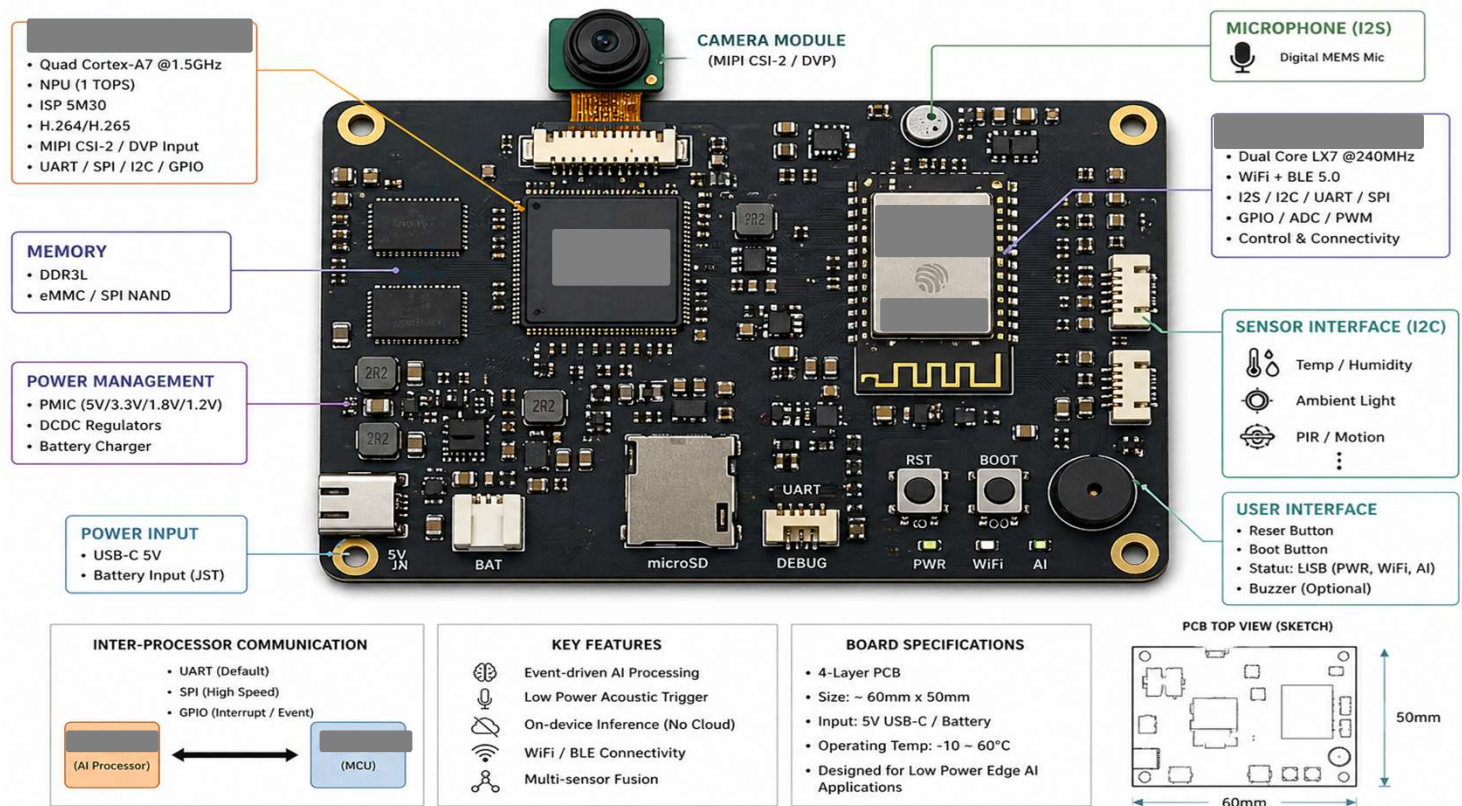
→ Event-driven Edge AI architecture defined (AI co-processor + MCU)

“Experimental results directly inform system architecture design.”



Validated under real-world noise conditions using FFT spectrum analysis

Edge AI - Intelligence Node Design



“Event-driven architecture enables real-world deployment at scale — where power, cost, and reliability define success.”